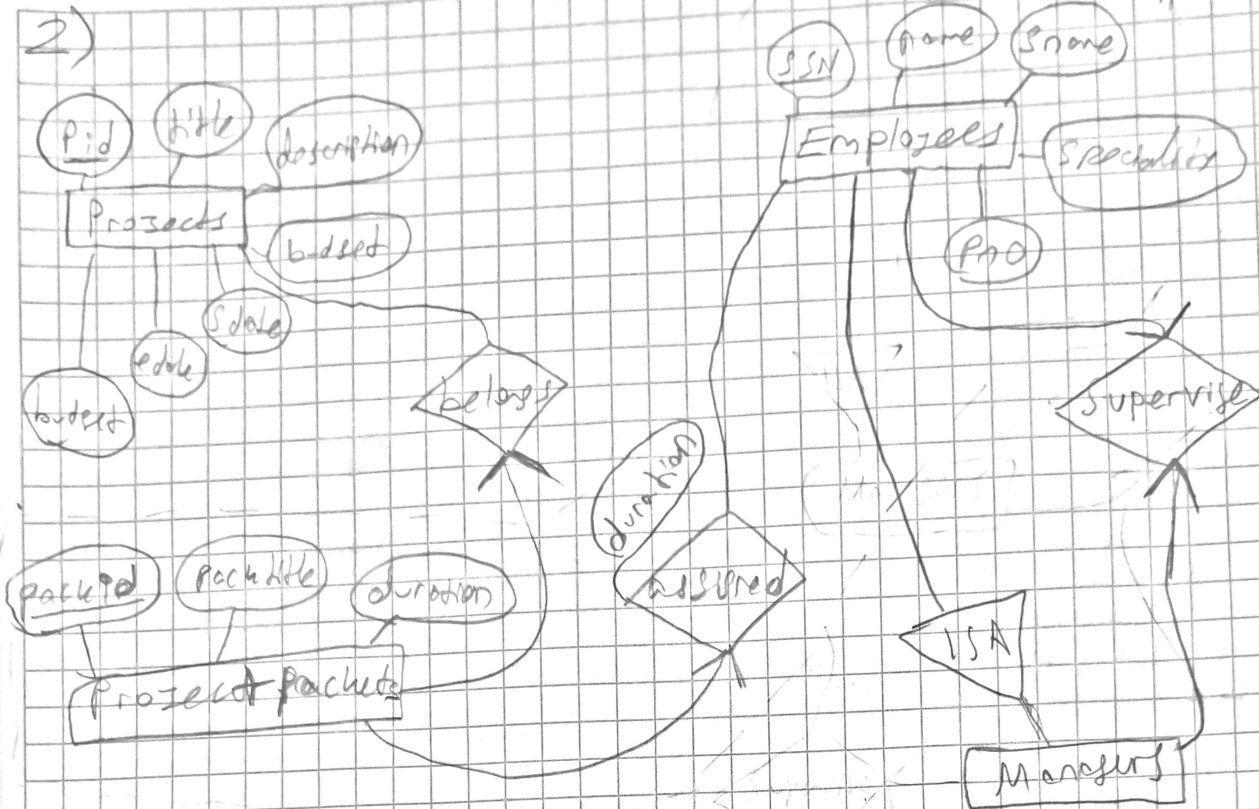


Subject :

2)



3) PATIENTS(SSN, Name, Phone)

DRUGS (Barcode, Name, Description)

DOCTORS(SSN, Name, Specialty)

Prescription(SSN, Barcode, Amount, Date)

CREATE TABLE Prescription(

CREATE TABLE Patients (SSN CHAR(10),
name CHAR(30),
PRIMARY KEY(SSN))

CREATE TABLE Drugs (Barcode CHAR(10),
name CHAR(30),
PRIMARY KEY(Barcode))

CREATE TABLE Doctors (SSN CHAR(10),
name CHAR(30),
PRIMARY KEY(SSN))

CREATE TABLE Prescription(
SSN CHAR(10),
Barcode CHAR(10),
Amount INTEGER,
Date Date,
PRIMARY KEY(SSN, Barcode)
FOREIGN KEY(SSN)
REFERENCES Patients
ON DELETE CASCADE,
FOREIGN KEY(Barcode)
REFERENCES Drugs)

Cohit Barkun Bze

2018556058

Subject :

Date :/...../.....

1) General external merge sort # of merge file (11)
3-mer pass (11)

a) Number of passes : $1 + \lceil \log_{B-1} \lceil N/B \rceil \rceil$

Cost = $2N * (\# \text{ of passes})$

Number of
pages in file
(N) = $\frac{150000}{(20/3)} = 22500$

1 page can
store = $\frac{B}{R} = \frac{1000}{300} = \frac{20}{3}$

Number of passes = $1 + \lceil \log_{10} \lceil 22500/11 \rceil \rceil = 5$

↓
 $\lceil 3.311 \rceil = 4$

Cost = $2 * 22500 * 5 = 22500$

Time required = $s + r + b * b/t = 10 + 7 + (5 * 0.5) = 12 + 2.5 = 14.5 \text{ ms}$

b)

Time = $5 * (12.5) = 62.5$